

ALTOS® Loose Tube, Gel-Free Cable 12 F, SMF-28® ULL, Single-mode (OS2)

CORNING

Part Number:
012PU4-T4119D20

Corning ALTOS® all-dielectric gel-free cables are designed for outdoor and limited indoor use for backbones in lashed aerial and duct installations. The loose tube gel-free design is fully waterblocked using craft-friendly, water-swappable materials, which means cable access is simple and no clean up is required. The flexible craft-friendly buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy mid-span access. The all-dielectric cable construction requires no bonding or grounding, and these cables have a medium-density polyethylene jacket that is rugged, durable and easy to strip.

Features and Benefits

Gel-free waterblocking technology

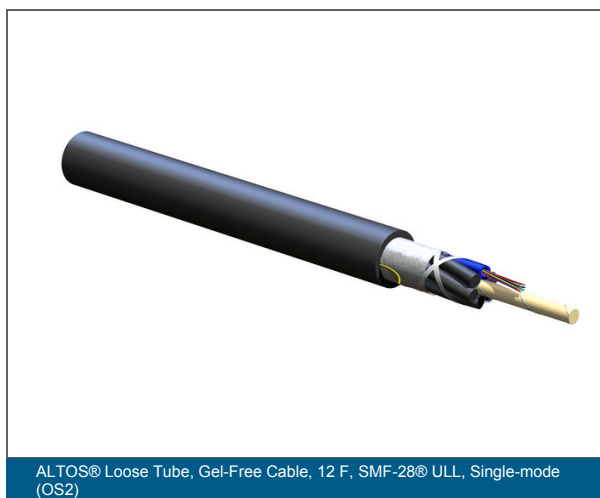
Craft-friendly cable preparation

Polyethylene jacket

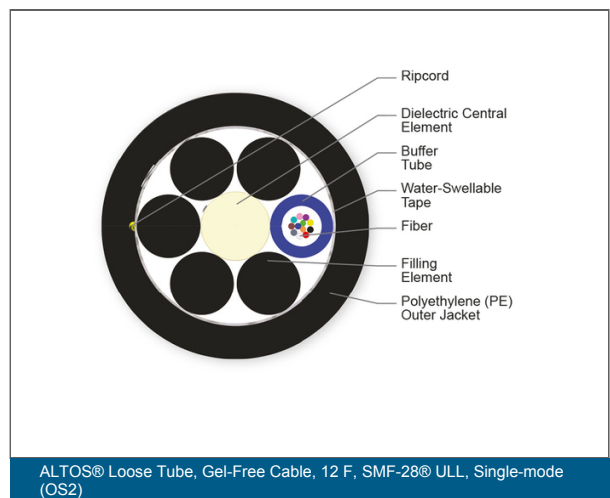
Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

All-dielectric construction

Requires no grounding or bonding



ALTOS® Loose Tube, Gel-Free Cable, 12 F, SMF-28® ULL, Single-mode (OS2)



ALTOS® Loose Tube, Gel-Free Cable, 12 F, SMF-28® ULL, Single-mode (OS2)

ALTOS® Loose Tube, Gel-Free Cable 12 F, SMF-28® ULL, Single-mode (OS2)

CORNING

Specifications

General Specifications	
Installation Methods	Aerial, Duct
Cable Type	Loose Tube
Environment	Outdoor
Product Type	Dielectric
Fiber Category	Ultra Low-Loss, Single-mode (OS2)
Cable geometry	Round

Standards	
RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Common Installations	Outdoor lashed aerial and duct, indoor when installed according to National Electrical Code® (NEC®) Article 770
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions	
Notes	Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results.
Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-40 °C to 70 °C (-40 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design	
Central Element	Dielectric
Fiber Count	12
Number of Ripcords	1
Outer Jacket Color	Black
Outer Jacket Material	Polyethylene (PE)

ALTOS® Loose Tube, Gel-Free Cable 12 F, SMF-28® ULL, Single-mode (OS2)

CORNING

Cable Design

Buffer Tube Color	Blue
Buffer Tube Diameter	2.5 mm (0.1 in)
Number of Active Tubes	1
Number of Filling Elements	5
Number of Tube Positions	6
Tape	Water-swellable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12
Color Code Standards	Telcordia

Mechanical Specifications

Max. Tensile Strength, Long-Term	890 N (200.08 lbf)
Max. Tensile Strength, Short-Term	2700 N (606.98 lbf)
Nominal Outer Diameter	10.5 mm (0.41 in)
Min. Bend Diameter Installation	316 mm (12.44 in)
Min. Bend Diameter Operation	210 mm (8.27 in)

Optical Characteristics

Fiber Code	P
Performance Option Code	19
Fiber Category	OS2
Fiber Type	Single-mode (OS2) / 250 µm
Fiber Name	SMF-28® ULL
Maximum Attenuation	0.33 dB/km / - / 0.19 dB/km
Wavelengths	1310 nm / 1383 nm / 1550 nm
Fiber Compliance	ITU-T G.654.C

**ALTOS® Loose Tube, Gel-Free Cable 12 F,
SMF-28® ULL, Single-mode (OS2)**



Dimensions	
Cable Weight	73 kg/km (49.05 lb/1000 ft)



Corning Optical Communications LLC • 4200 Corning Place • Charlotte, NC • 28216 • United States
800-743-2675 • FAX: 828-325-5060 • International: +1-828-901-5000 • www.corning.com/opcomm

A complete listing of the trademarks of Corning Optical Communications is available at www.corning.com/opcomm/trademarks. All other trademarks are the properties of their respective owners. Corning Optical Communications is ISO 9001 certified. © 2026 Corning Optical Communications. All rights reserved.